

Piyush Kumar Singh

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PROFESSIONAL SUMMARY

AI Engineer building production-grade Generative AI and ML systems, including Agentic RAG pipelines, fine-tuned LLMs, and real-time inference APIs deployed in industrial environments. GCP-certified with 5 years of experience across ML engineering and enterprise solutioning, combining strong technical execution with business understanding. Proven ability to take AI solutions from problem definition to production deployment, working closely with stakeholders to deliver scalable, high-impact systems.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, FAISS, Vector Databases, LoRA/PEFT

Machine Learning & Data: Scikit-learn, Pandas, NumPy, Feature Engineering, Model Training, Model Evaluation

Deep Learning: PyTorch, TensorFlow, NLP, CNNs, RNNs, Transformers

Languages: Python, SQL

Tools & Deployment: FastAPI, Docker, Streamlit, HuggingFace, Git, BeautifulSoup, Selenium, GCP

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | [VAMS Mechatronics Pvt. Ltd.](#)

Aug 2024 – Present

Ludhiana, Punjab

- Developed ML models for predictive maintenance across multiple industrial environments, using multivariate sensor data (vibration, temperature, current, RPM) to identify abnormal equipment behavior
- Performed EDA, data cleaning, and feature engineering on time-series data, improving data quality and enabling effective model training
- Trained and evaluated Isolation Forest and XGBoost models for equipment risk scoring and anomaly identification, contributing to ~25–30% reduction in unplanned downtime during pilot use cases
- Built a FastAPI-based inference API to serve model predictions, enabling maintenance teams to receive early warning signals (~24–48 hours prior) via internal dashboards

Senior Sales Engineer | [Quantiphi Analytics](#)

Apr 2021 – Jul 2024

Bangalore, Karnataka

- Owned pre-sales for GCP cloud migration and data warehouse modernization engagements, collaborating with architects on solution design and delivering proposals, SOWs, and RFPs for \$250K to \$2M deals across 20+ enterprise clients in US Central and LATAM
- Engaged client stakeholders through requirement gathering, solution presentations, and technical scoping, translating complex architecture decisions into clear business value across data infrastructure and cloud transformation projects
- Contributed to regional pre-sales portfolio growth from \$7M to \$20M over 3 years through consistent delivery and client engagement quality

PROJECTS

PDF Chatbot — Agentic RAG System | [Demo](#)

Python · LangChain · Gemini 2.5 Flash · FAISS · Vector Database · Streamlit · Docker · HuggingFace Spaces

- Built an Agentic Retrieval-Augmented Generation (RAG) system using LangChain, Gemini 2.5 Flash, and FAISS vector database — retrieval exposed as a LangChain Agent tool, enabling the LLM to invoke context fetch only when needed rather than on every query

- Engineered rate limit handling (TPM guard, 429 retry with exponential backoff) and optimised chunk size from 1,500 → 2,500 chars, reducing API consumption by 40% while maintaining retrieval quality
- Deployed on HuggingFace Spaces via Docker — publicly accessible with full PDF upload, natural language processing, and source-grounded Q&A

Valorant Champions 2024 — Esports Data Analysis | [Report](#)

Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter

- Built an end-to-end EDA and machine learning pipeline on a custom-scraped 11-file dataset analysing 16 teams and 80+ players across match, player, agent, map, and economy dimensions
- Engineered a composite player performance score across 6 weighted metrics (FK, FD, kills, deaths, ADR, APR) — model aligned with final tournament standings, validating predictive accuracy against real outcomes
- Quantified ADR–ACS correlation at 0.96; established pistol round control as primary win predictor — teams winning both pistols convert 90% of maps vs 10% when losing both

VLR.gg Esports Data Scraper | [Demo](#)

Python · Selenium · BeautifulSoup4 · Pandas · Streamlit · Git

- Built a modular Python scraping pipeline extracting Valorant tournament data from VLR.gg — capturing match results, player stats, agent usage, economy metrics, and Attack/Defense performance splits across complete event datasets
- Engineered a hybrid scraping architecture using BeautifulSoup4 for static pages and Selenium for dynamically rendered match data, coordinating 6 scraper modules through a central orchestration layer
- Developed a Streamlit interface enabling a scrape, validate, and export workflow with real-time progress tracking and export of 10+ structured datasets in JSON and CSV formats.
- Open-source contribution — 9 published Kaggle datasets, 1,600+ downloads, 60+ upvotes

Stable Diffusion 1.5 — LoRA Fine-tuning

Python · KohyaSS · ComfyUI · Stable Diffusion 1.5 · LoRA

- Fine-tuned Stable Diffusion 1.5 using LoRA (KohyaSS) for consistent character generation — managing dataset curation, preprocessing, and augmentation pipeline
- Built ComfyUI workflows for controlled generative outputs, evaluating style and character consistency across prompt variations.

EDUCATION

B.Tech — Computer Science

Lovely Professional University, Jalandhar

Aug 2017 – Jul 2021

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate — Oracle

Feb 2026

Deep Learning Specialization — DeepLearning.AI — Andrew Ng

Jan 2025

Associate Cloud Engineer — Google Cloud Platform (GCP)

Feb 2022